



PARCNet: Phase-aware residual correction network for efficient multivariate time series forecasting

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ABSTRACT

In multivariate time series forecasting, incorporating oscillation-related representations has attracted increasing attention. However, most existing studies primarily emphasize amplitude-based modeling, while overlooking phase information. Under strong non-stationary disturbances, this omission may cause feature misalignment and unreliable fusion, leading to amplified error accumulation in long-horizon forecasting. To address this issue, we propose PARCNet (Phase-Aware Residual Correction Network), a framework that explicitly models envelope and phase to characterize local oscillatory states and improve long-horizon forecasting stability. Specifically, PARCNet adopts a Hilbert analytic-signal perspective to transform real-valued sequences into amplitude–phase descriptors aligned in time. A lightweight one-dimensional convolutional gating mechanism then produces time-channel adaptive weights, which drive a phase-aware residual correction to refine features in a structure-consistent manner. This design enhances representational expressiveness, corrects local oscillatory deviations, and improves the temporal consistency of cross-variable coupling. Moreover, PARCNet is implemented as a plug-and-play module that can be seamlessly integrated into existing forecasting backbones to strengthen multivariate modeling capability. Extensive experiments on twelve real-world datasets demonstrate that PARCNet consistently outperforms state-of-the-art methods, achieving higher accuracy and robustness across diverse prediction horizons and evaluation settings. The code and datasets are available at <https://github.com/9527dandelion/PARCNet>.

1. Introduction

Multivariate time series forecasting plays a critical role in areas such as weather prediction, energy dispatch, traffic monitoring, and financial risk management [1,2]. In recent years, deep learning methods have substantially improved the ability to model complex temporal dependencies and cross-variable interactions [3,4], with Transformer-based approaches excelling at capturing long-range dependencies and Multi-Layer Perceptron (MLP) based approaches demonstrating high modeling efficiency [5]. Overall, mainstream deep forecasting models primarily operate in the time domain, relying on end-to-end learning to capture dynamics across both time and variables [6,7].

However, real-world sequences often exhibit non-stationary drifts and local oscillatory patterns, which can cause time-domain models to accumulate errors and produce cross-variable misalignment in long-horizon predictions. To address this, researchers have recently incorporated frequency-domain modeling [4,8], explicitly capturing periodic structures and energy distributions to enhance multi-scale

pattern extraction. Techniques such as frequency-domain filtering, sub-band normalization, and frequency-based periodicity extraction have been proposed to improve the stability and efficiency of sequence representations [9,10]. While these frequency-domain methods achieve notable success in modeling periodicity and multi-scale features, most still rely primarily on amplitude or energy information, lacking direct and stable modeling of phase synchronization, phase drift, and local coupling in multivariate oscillatory systems [11,12]. Consequently, models remain prone to cross-variable misalignment and amplified prediction errors in long-horizon forecasting [13,14].

It should be noted that the proposed PARCNet is not merely a phase-enhanced variant of existing frequency-domain methods. Most existing spectral-domain methods mainly exploit frequency magnitude, energy distribution, frequency filtering, or selected frequency components for prediction. In contrast, PARCNet adopts the Hilbert analytic-signal perspective to obtain time-aligned instantaneous envelope and phase descriptors, which preserve local oscillatory states in the original temporal coordinate. More importantly, these amplitude–phase

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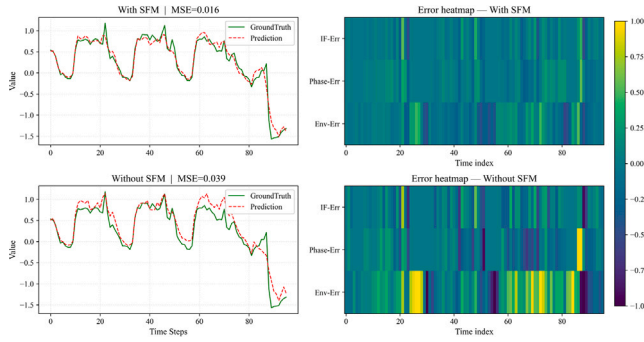


Fig. 1. Ablation comparison of SFM on the Electricity Load (ECL) dataset (Horizon = 96). The left panels plot the ground-truth versus predicted values for the selected variable, and the right panels visualize errors in three analytic-signal descriptors: instantaneous frequency (IF-Err), phase (Phase-Err), and envelope (Env-Err). Compared with the model without SFM (bottom row), the model with SFM (top row) exhibits smaller errors, especially around rapid local fluctuations, indicating improved modeling of local oscillatory structures.

descriptors are not simply concatenated as auxiliary features. They are used to generate time-channel adaptive gates, which control the residual correction from the data-driven representation toward a learnable structural prior. Therefore, the central contribution of PARCNet lies not in phase extraction alone, but in a phase-aware, structure-guided gated residual correction mechanism.

To address the aforementioned challenges, we propose **Phase-Aware Residual Correction Network (PARCNet)**, a forecasting model for multivariate time series that explicitly leverages amplitude and phase features to capture local oscillatory states in the sequence [15], thereby enhancing long-horizon prediction stability and reliability [16,17]. Its core **Structure-aware Feature Module (SFM)** extracts structured representations of oscillation intensity as well as phase synchronization and phase drift across variables, providing stable and interpretable features for downstream prediction. Unlike conventional models that rely solely on amplitude, energy, or time-domain representations, PARCNet utilizes explicit amplitude and phase information to describe local oscillatory states. However, phase information is not treated as an isolated forecasting feature in PARCNet. Instead, it serves as a structural condition for adaptive feature correction. Specifically, PARCNet applies the Hilbert transform to construct an analytic-signal representation of the input sequence, from which the instantaneous envelope (amplitude) and instantaneous phase are derived to characterize local oscillatory states [18,19]. These analytic-signal-based amplitude-phase descriptors are then stacked and fed into a one-dimensional convolution to produce a time-channel adaptive gating signal. This gate further controls the residual correction from the data-driven representation toward a learnable structural prior, thereby forming a phase-aware and structure-consistent fused representation. Finally, the fused representation is further modeled via self-attention and MLP modules to capture complex temporal dependencies and cross-variable interactions, yielding the final prediction. As shown in Fig. 1, the proposed structure-aware residual fusion representation significantly improves prediction accuracy. The corresponding error heatmap indicates a notable reduction in errors in envelope, phase, and instantaneous frequency, demonstrating improved modeling of local oscillatory features. In contrast, the predictions without this design exhibit larger errors in regions of local fluctuations. This design effectively enhances the model's ability to perceive local oscillatory features, thereby improving the stability and reliability of multivariate time series forecasting over long prediction horizons.

In summary, the main contributions of this work are as follows:

- We propose **PARCNet**, a phase-aware residual correction framework for multivariate time series forecasting. Different from conventional frequency-domain methods that mainly exploit spectral

magnitude or energy information, PARCNet uses Hilbert analytic-signal descriptors to characterize time-aligned local oscillatory states, including envelope variation and phase evolution;

- We design a **Structure-aware Feature Module (SFM)**, where amplitude-phase descriptors are not simply used as auxiliary inputs but are transformed into time-channel adaptive gates. These gates guide the residual correction from the data-driven representation toward a learnable structural prior, enabling structure-consistent feature refinement and mitigating local oscillatory misalignment;
- We conduct extensive experiments on twelve real-world datasets. The results demonstrate that PARCNet outperforms state-of-the-art methods in both accuracy and stability: among 120 different prediction lengths and evaluation settings, PARCNet ranks first in 98 cases and within the top two in 115 cases.

2. Related work

2.1. Deep learning methods for time series forecasting

In recent years, deep learning methods have achieved significant progress in time series forecasting.

Multi-layer perceptron (MLP)-based approaches have attracted widespread attention due to their structural simplicity and computational efficiency [5,20]. DLinear combines trend-seasonal decomposition with linear modeling, achieving competitive performance under low complexity, while RLinear [21] further enhances representation capacity through residual modeling. For multivariate forecasting, TimeMixer [22] and its improved variant TimeMixer++ [23] enable cross-dimensional information interaction by mixing features along both temporal and variable dimensions.

Convolutional neural networks (CNNs) model time series through local receptive fields [24]. TimesNet [25] employs multi-scale convolutional structures to capture periodic patterns at different temporal granularities [26]. ModernTCN enhances long-sequence modeling via deep temporal convolutions [27], while SCINet [28] leverages hierarchical decomposition and recursive convolutional architectures to effectively characterize complex temporal dependencies.

Transformer-based methods demonstrate strong capability in modeling long-range dependencies through self-attention mechanisms [29,30]. Informer and Reformer reduce computational complexity by adopting sparse or approximate attention [31], enabling Transformer models to scale to long-horizon forecasting tasks. Autoformer replaces standard self-attention with an autocorrelation-based mechanism to better capture periodic structures [8]. iTransformer [32] applies attention across variables to strengthen cross-variable interaction modeling, while TQNet [33] uses periodically shifted Temporal Queries to capture global inter-variable patterns efficiently.

Overall, these deep learning methods have improved time series forecasting from different perspectives, including efficient temporal modeling, local pattern extraction, long-range dependency learning, and cross-variable interaction modeling. However, most of them primarily operate on time-domain representations or learned latent features, and the local oscillatory structure of multivariate sequences is usually captured implicitly. In particular, phase evolution, phase synchronization, and phase-related local misalignment are rarely modeled as explicit structural factors. This motivates us to introduce amplitude-phase descriptors into the forecasting framework and further use them for structure-aware feature correction.

2.2. Frequency-domain methods for time series forecasting

In recent years, frequency-domain information has attracted increasing attention in time series forecasting, with related studies primarily focusing on incorporating spectral features into deep learning models to better capture periodic patterns and multi-scale structures [34].

Transformer-based approaches typically integrate frequency-domain modeling with self-attention mechanisms. FEDformer performs feature representation and modeling in the frequency domain [9], fusing time- and frequency-domain information to enhance multi-scale forecasting performance. FredFormer further introduces frequency band partitioning and sub-band normalization to alleviate frequency bias and achieve more balanced spectral modeling [14].

MLP-based methods emphasize improving modeling efficiency and stability through frequency-domain transformations [35]. FreTS incorporates frequency-domain representations to model both temporal dependencies and inter-variable interactions in multivariate sequences [36]. FilterNet employs learnable frequency filters to extract salient spectral components [13], while Amplifier enhances low-energy frequency components to improve predictive accuracy [37].

Overall, existing frequency-domain methods mainly focus on frequency magnitude, energy distribution, frequency filtering, or frequency-band selection, with limited explicit modeling of phase evolution and local oscillatory states. Different from these spectral-domain approaches, PARCNet does not aim to perform forecasting directly in the transformed frequency domain, nor does it simply concatenate phase as an auxiliary feature. Instead, PARCNet adopts the Hilbert analytic-signal perspective to obtain time-aligned envelope and phase descriptors, and further uses these amplitude-phase descriptors to generate adaptive gates. These gates guide the residual correction from the data-driven representation toward a learnable structural prior, enabling structure-aware correction of local oscillatory misalignment. Therefore, the key distinction of PARCNet lies not in phase extraction alone, but in phase-aware, structure-guided gated residual correction beyond conventional magnitude-oriented spectral modeling.

3. Problem statement

This section mainly defines the fundamental concepts and notations of time series and formally describes the forecasting problem studied in this work.

Definition 1 (Multivariate Time Series Forecasting). The goal of multivariate time series forecasting is to estimate future values of multiple variables based on past observations. Let the historical observation sequence be:

$$X = \{x_1, x_2, \dots, x_L\} \in \mathbb{R}^{C \times L}, \quad (1)$$

where $x_t \in \mathbb{R}^C$ represents the values of C variables at time step t , and L is the length of the look-back window. The prediction target is the future sequence of length H :

$$Y = \{y_1, y_2, \dots, y_H\} \in \mathbb{R}^{C \times H}, \quad (2)$$

where $y_t \in \mathbb{R}^C$ denotes the values of the C variables at future time step t . The objective is to learn a mapping model f that transforms the historical sequence X into the predicted future sequence $\hat{Y}$:

$$\hat{Y} = F(X). \quad (3)$$

Definition 2 (Hilbert Transform and Analytic Signal). The Hilbert transform is a linear operator that maps a real-valued time series to a signal orthogonal to the original one, enabling the extraction of instantaneous phase information. Let a real-valued time series of length T be:

$$X = \{x_1, x_2, \dots, x_T\}. \quad (4)$$

The Hilbert transform $\mathcal{H}(X)$ is defined as [38]:

$$\mathcal{H}(X)(t) = \frac{1}{\pi} \text{p.v.} \int_{-\infty}^{\infty} \frac{X(\tau)}{t - \tau} d\tau, \quad (5)$$

where p.v. denotes the Cauchy principal value. Based on this, the analytic signal is defined as:

$$X_H = X + i \mathcal{H}(X), \quad (6)$$

where X is the real part, $\mathcal{H}(X)$ is the imaginary part, and $i = \sqrt{-1}$. The analytic signal X_H contains instantaneous phase information, which can be expressed as [39]:

$$\phi(t) = \arg(X_H(t)) = \arctan \frac{\mathcal{H}(X)(t)}{X(t)}. \quad (7)$$

It should be noted that the phase obtained from the Hilbert analytic signal is a mathematical analytic-signal phase. For strongly non-stationary or multi-component signals, its direct physical interpretation may become ambiguous, especially when different oscillatory components are heavily mixed or their frequency bands strongly overlap. In such cases, the classical narrow-band assumption and the conditions related to Bedrosian's theorem may not be strictly satisfied. Therefore, PARCNet does not rely on the assumption that the Hilbert-derived phase is a strictly interpretable physical phase for all signal conditions. Instead, the Hilbert-derived envelope and phase are used as data-driven descriptors of local oscillatory states, providing structural information for adaptive feature gating and residual correction.

4. PARCNet

4.1. The overall architecture

Given a historical multivariate time series $X = \{x_1, x_2, \dots, x_L\} \in \mathbb{R}^{C \times L}$, where C denotes the number of variables (channels) and L the sequence length, the goal of multivariate time series forecasting is to predict the future sequence over a horizon H : $Y = \{x_{L+1}, x_{L+2}, \dots, x_{L+H}\} \in \mathbb{R}^{C \times H}$.

As shown in Fig. 2, we propose **PARCNet**, a Hilbert amplitude-phase based enhancement framework for multivariate time series forecasting. Given an input multivariate time series X , the model first applies *Instance Normalization* to perform instance-wise normalization, which helps alleviate non-stationarity and stabilize the training process. The normalized sequence is then projected into a high-dimensional feature space through an embedding module to enhance representation capacity. Built upon the embedded representation, PARCNet introduces a *Structure-aware Feature Module* (SFM) to explicitly model structural oscillatory information in the sequence. Specifically, the module jointly considers the embedded input sequence and a *learnable structural prior* T as conditioning signals. By applying the Hilbert transform, analytic signals are constructed for both components, from which amplitude and phase features are extracted. These amplitude-phase features are subsequently stacked and fused via one-dimensional convolution, producing a time-channel adaptive gating signal g . Based on this gating signal, a structure-aware residual representation f is constructed to selectively correct the original features in a structure-consistent manner. Finally, the refined representation is fed into a multi-head attention module to capture complex temporal dependencies and cross-variable interactions. The resulting output is then passed through a linear projection followed by inverse normalization to produce the final prediction Y .

4.2. Structure-aware Feature Module (SFM)

The Structure-aware Feature Module (SFM) is designed to explicitly model structural oscillatory information in multivariate time series and to adaptively refine feature representations in a structure-aware manner. Specifically, SFM consists of three sequentially connected submodules. The *Structural Oscillation Representation* (SOR) introduces the structural prior and characterizes local oscillatory states of the sequence via analytic signal representations. The *Spectral Feature Gating Mechanism* generates time-channel adaptive gating signals based on amplitude-phase features [15], enabling selective modulation of spectral structural information. Finally, the *Structure-aware Residual Correction* leverages the gating results to perform structure-aware residual refinement of the original features, thereby enhancing the modeling of structural oscillatory patterns while preserving representation stability.

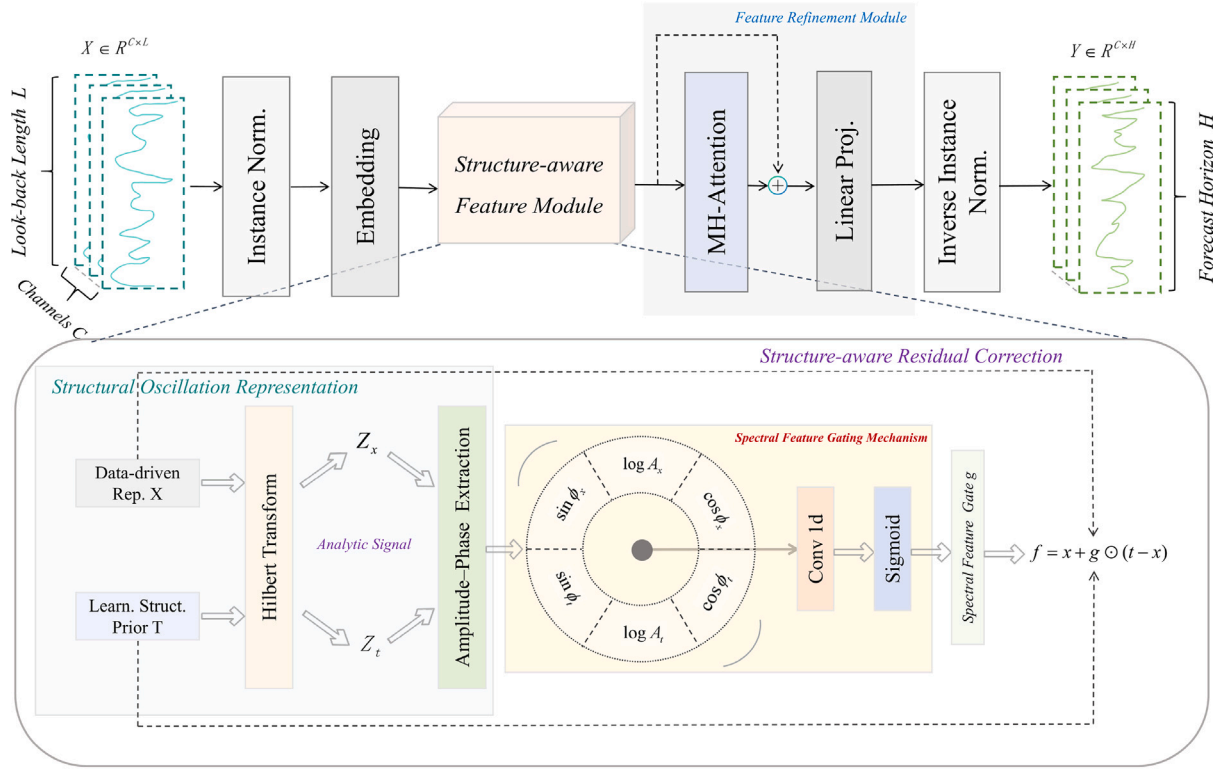


Fig. 2. Overall architecture of **PARCNet** for multivariate time series forecasting. The input sequence $X \in \mathbb{R}^{C \times L}$ is normalized and embedded, followed by refinement through the Structure-aware Feature Module (SFM). The refined features are further processed by a feature refinement module composed of multi-head attention and linear projection, and inverse instance normalization is applied to produce the prediction $Y \in \mathbb{R}^{C \times H}$. The lower panel illustrates the internal workflow of the SFM.

Structural Oscillation Representation (SOR). The SOR aims to construct structured representations of oscillatory states in multivariate time series, providing a stable structural basis for subsequent structure selection and residual correction. This module is conditioned on two complementary sources of information: (i) a fully data-driven feature representation X obtained from the embedding module, which characterizes the sample-specific observed patterns at the current temporal scale; and (ii) a learnable structural prior T , which serves as a sample-agnostic trainable reference in the embedded feature space.

Specifically, T is not a manually predefined template or an external physical prior. Instead, it is implemented as a trainable parameter shared by all input samples within the same dataset and optimized jointly with the other network parameters. In our implementation, T has the same time-channel resolution as the representation processed by SFM, i.e., $T \in \mathbb{R}^{1 \times C \times S}$, where C denotes the number of variables and S denotes the feature length after embedding. During the forward pass, T is broadcast along the batch dimension to match the shape of X . Since T is learned only from the training objective and does not use any future target values during inference, it does not introduce information leakage.

The rationale behind introducing T is that multivariate time series often contain shared latent oscillatory structures across samples, such as common envelope variations, phase evolution tendencies, and cross-variable synchronization patterns. The data-driven representation X captures the local state of each input sample, but it may be affected by non-stationary disturbances, noise, or local phase shifts. In contrast, T provides a stable structural reference learned from the training distribution. By jointly considering X and T , SOR enables the model to compare sample-specific oscillatory states with a learned structural reference, while the subsequent gate g adaptively determines whether and to what extent the original representation should be corrected toward this reference.

Based on the above inputs, SOR applies the Hilbert transform to both the data-driven representation X and the learnable structural prior T to construct their corresponding analytic signal representations. Specifically, the analytic forms are defined as:

$$Z_x = X + i \mathcal{H}(X), \quad Z_t = T + i \mathcal{H}(T), \quad (8)$$

where $\mathcal{H}(\cdot)$ denotes the Hilbert transform and i is the imaginary unit. The analytic signal provides a unified complex-valued representation that simultaneously encodes the local oscillation amplitude and phase.

Given the analytic signals Z_x and Z_t , the corresponding amplitude and phase components are extracted. The amplitude is obtained as the modulus of the complex signal:

$$A_x = |Z_x|, \quad A_t = |Z_t|. \quad (9)$$

To mitigate scale variations across variables and time steps and to improve numerical stability, a logarithmic transformation is applied to the amplitude terms:

$$\tilde{A}_x = \log(A_x + \epsilon), \quad \tilde{A}_t = \log(A_t + \epsilon), \quad (10)$$

where ϵ is a small constant introduced to avoid numerical instability.

The phase information is given by the argument of the analytic signal. To avoid discontinuities inherent in the raw phase angle, we adopt an explicit sine-cosine representation. Let

$$\phi_x = \arg(Z_x), \quad \phi_t = \arg(Z_t), \quad (11)$$

then the phase features are represented as

$$P_x = [\cos(\phi_x), \sin(\phi_x)], \quad P_t = [\cos(\phi_t), \sin(\phi_t)]. \quad (12)$$

Through the above formulation, SOR maps both the data-driven representation and the structural prior into a unified amplitude-phase feature space, enabling a structured characterization of local oscillatory

states without relying on explicit frequency assumptions. This representation serves as a stable foundation for subsequent spectral feature gating and structure-aware residual correction.

To improve robustness on real-world non-stationary sequences, PARCNet does not directly use the raw phase angle as an independent prediction target. Instead, the phase is represented by sine-cosine components to avoid discontinuities caused by phase wrapping, and it is combined with logarithmic amplitude features to form a multi-attribute oscillatory descriptor. The subsequent convolutional gate further learns when and where such amplitude-phase information should be emphasized. In this way, potential imperfections of Hilbert-derived phase under noisy or multi-component conditions are mitigated by adaptive gating and structure-aware residual correction rather than being directly propagated to the final prediction.

Spectral Feature Gating Mechanism (SFGM). The Spectral Feature Gating Mechanism is designed to adaptively select and modulate spectral information based on structured amplitude-phase representations, thereby generating structure-aware gating weights. This module takes the multi-source amplitude-phase features produced by the SOR as input and jointly models different oscillatory attributes to learn time-channel adaptive gating responses, which are used to modulate the strength of spectral feature activations under varying structural states.

Specifically, the SFGM takes the six-dimensional analytic-signal domain amplitude-phase features output by the SOR as input. Based on the structured oscillatory representations constructed in the previous section, the model organizes the logarithmic amplitude features and the sine-cosine representation of phase into the following spectral feature set:

$$F = [\tilde{A}_x, \tilde{A}_t, \cos(\phi_x), \sin(\phi_x), \cos(\phi_t), \sin(\phi_t)] \in \mathbb{R}^{(B \cdot C) \times 6 \times S}, \quad (13)$$

where B denotes the batch size, C the number of channels, and S the sequence length after embedding. The six features are stacked along the feature-type dimension to form a unified amplitude-phase representation, capturing multi-attribute structural information under local oscillatory states.

To efficiently capture the local dependencies of amplitude-phase descriptors along the sequence dimension, the model employs a two-layer one-dimensional convolutional network to map the stacked amplitude-phase descriptors:

$$g = \sigma(\text{Conv}_2(\text{Conv}_1(F))) \in \mathbb{R}^{B \times C \times S}, \quad (14)$$

where $\text{Conv}_1(\cdot)$ and $\text{Conv}_2(\cdot)$ denote one-dimensional convolution operators, and $\sigma(\cdot)$ is the sigmoid gating function. The resulting g serves as a time-channel adaptive spectral feature gate, modulating the amplitude-phase descriptors under different structural states and providing a key control signal for the subsequent structure-aware residual correction.

In summary, SFGM organizes and stacks the amplitude and phase information of structured oscillatory features to construct a multi-attribute amplitude-phase representation. The two-layer convolutional gating network then adaptively generates time-channel modulation weights, enabling fine-grained control over amplitude-phase descriptors under different local oscillatory states. This mechanism not only enhances the expression of important salient oscillatory patterns but also provides crucial control signals for structure-aware residual correction, thereby improving the model's capability to capture complex structures in multivariate time series.

Structure-aware Residual Correction (SRC). In multivariate time series, even after amplitude-phase feature modeling and spectral gating, local oscillatory states across different channels may still exhibit shifts or inconsistencies. Such shifts can cause deviations in the original features at a fine-grained structural level, which may negatively affect prediction stability and accuracy. To address this, the SRC module introduces a *structure-aware residual* mechanism to correct the original features, thereby enhancing the model's adaptability to local oscillatory patterns.

Let X denote the embedded original features, T the learnable structural prior, and g the time-channel adaptive gating signal generated from the spectral amplitude-phase features. The output features f of SRC are defined as:

$$f = X + g \odot (T - X), \quad (15)$$

where $\odot$ represents element-wise multiplication. This formulation can be interpreted as a phase-aware gated residual correction from the data-driven representation X toward the learnable structural prior T . It should be emphasized that SRC does not explicitly shift the raw phase angle or perform a manually designed phase-alignment operation. Instead, the phase-aware property comes from the generation of the gate g . As defined in the amplitude-phase feature set and the subsequent convolutional gating function, g is inferred from the amplitude-phase descriptors of both X and T , including logarithmic envelope features and sine-cosine phase features. Therefore, the model can adaptively assign larger correction weights to time-channel positions where the local oscillatory state of X is inconsistent with the learned structural reference T , and assign smaller weights where the original representation is already structurally reliable.

From this perspective, the update $X + g \odot (T - X)$ is not a generic linear interpolation, but an amplitude-phase-conditioned residual correction. The direction $T - X$ represents the structural discrepancy between the sample-specific representation and the learned structural reference, while g controls how strongly this discrepancy should be corrected under different local oscillatory states. When phase drift or local oscillatory inconsistency causes feature misalignment, the amplitude-phase-conditioned gate increases the contribution of the structural prior, moving the representation toward a more stable structural reference. When the local structure is stable, the gate remains small and the original representation is preserved. In this way, SRC mitigates phase-related feature misalignment at the representation level rather than directly manipulating the raw phase angle.

As shown in Fig. 3, the structure-aware residual correction (SRC) process can be intuitively represented in a three-dimensional coordinate system. In the figure, the line segment s denotes the original feature X , and the line segment t denotes the structural prior T . Under different gating weights g , the original feature shifts toward the structural prior to varying extents, corresponding to the line segments f_1 and f_2 , respectively. When g is large, the feature moves closer to the structural prior (as in f_1); when g is small, the shift is minor, and the original feature remains relatively stable (as in f_2). This visualization provides an intuitive illustration of how the SRC module dynamically adjusts the original feature based on the gating signal, achieving fine-grained structure-aware correction.

Through this structure-aware residual design, the SRC module enhances the consistency of local oscillatory patterns across different channels while preserving the expressive capacity of the original features. This approach effectively mitigates inter-channel shifts and inconsistencies, stabilizes gradient propagation, and avoids excessive dependence on the structural prior, thereby improving the robustness and representation capability of the model in multivariate time series forecasting.

4.3. Feature refinement module

The Feature Refinement Module (FRM) is designed to further process the multivariate time series features after structure-aware correction, capturing complex dependencies within the sequence and producing the final predictive representation. FRM mainly consists of two submodules: Multi-Head Attention (MHA) [29], which models interactions across time steps and channels, and Linear Projection, which maps the refined features to the prediction space, directly supporting the generation of the model's forecast output.

MHA. After obtaining the feature representation $f \in \mathbb{R}^{B \times S \times d}$, the Feature Refinement Module (FRM) applies multi-head self-attention to

capture dependencies along the temporal dimension. For each head, the attention is computed as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (16)$$

where the query, key, and value matrices are obtained via linear projections:

$$Q = fW_Q, \quad K = fW_K, \quad V = fW_V, \quad (17)$$

with $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$ being learnable projection matrices, and the projections are applied to the last (feature) dimension of f . This mechanism adaptively aggregates information across time steps, enabling the model to capture complex temporal dependencies and produce refined feature representations for the subsequent linear projection module.

Linear Projection. After refinement by the MHA, the features are mapped to the prediction space through a linear projection while incorporating a residual connection to enhance training stability [40]. The output of the linear projection is defined as:

$$\tilde{Y} = \text{Linear}(\text{Dropout}(\text{MHA}(f) + f)). \quad (18)$$

where $\text{MHA}(\cdot)$ denotes the multi-head attention block, f denotes the structure-aware corrected input features, and $\text{Dropout}(\cdot)$ is applied for regularization. This residual design preserves the original information while efficiently mapping the attention-refined features to the prediction space, enabling stable and accurate forecasting outputs.

The final output $\tilde{Y}$ is restored to the original scale through inverse instance normalization [41], producing the final predicted sequence:

$$Y = \text{IN}^{-1}(\tilde{Y}). \quad (19)$$

5. Experiments

5.1. Experimental settings

Dataset. We conduct a comprehensive evaluation of the proposed PARCNet model on multiple representative multivariate time series datasets. Specifically, several benchmark datasets widely used for general time series forecasting are adopted, including ETTh1, ETTh2, ETTm1, ETTm2, Electricity Load (ECL), Exchange, Weather, and Solar-Energy [31]. These datasets differ significantly in terms of variable dimensionality, sampling frequency, sequence length, and prediction horizons. Detailed dataset statistics as well as the corresponding train, validation, and test splits are summarized in Table 1. Standard pre-processing procedures provided with each dataset are followed in all experiments.

In addition, to further assess the model's capability in modeling high-dimensional traffic time series, four multivariate datasets from the traffic domain are incorporated, namely PEMS03, PEMS04, PEMS07, and PEMS08 [35]. These datasets are collected from the Performance Measurement System (PEMS) operated by the California Department of Transportation, with a temporal resolution of 5 min. Considering the characteristics of traffic data, prediction horizons different from those used in general forecasting benchmarks are employed. The dataset scale, variable dimensionality, and data split configurations are also reported in Table 2.

Baselines. To evaluate the performance of PARCNet, we select state-of-the-art baseline models for comparison according to the characteristics of different datasets. For the first eight general-purpose forecasting datasets, we include TQNet [33], TimePro [42], Amplifier [37], iTransformer [32], SOFTS [43], Fredformer [14], TiDE [44], TimesNet [26], and PatchTST [45] as representative baselines. For the four traffic datasets, which exhibit pronounced periodic patterns and spatial structures, we adopt TimeMixer [22], TimeMixer++ [23], FilterNet [13], FITS [46], CARD [47], Fredformer [14], Crossformer [30],

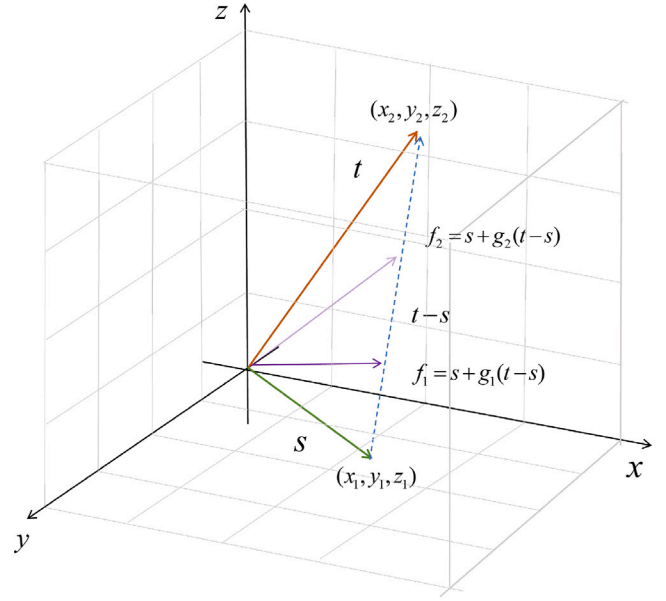


Fig. 3. Geometric interpretation of the proposed Structure-aware Residual Correction (SRC). The original feature s is corrected toward the structural prior t along the discrepancy direction $(t - s)$ with gating weight g , yielding $f = s + g \odot (t - s)$. This process illustrates a feature-space residual correction controlled by amplitude-phase-conditioned gates, rather than a direct manual shift of the raw phase angle.

Table 1

Statistics of the general-purpose forecasting benchmarks, including dimensionality, sampling frequency, sequence length, train/validation/test split, and prediction horizons.

Dataset	Dim	Frequency	Total len.	Split	Prediction len.	Information
ETTh1, ETTh2	7	Hourly	17 420	6: 2: 2	{96,192,336,720}	Electricity
ETTh1, ETTm2	7	15 min	69 680	6: 2: 2	{96,192,336,720}	Electricity
Weather	21	10 min	52 696	7: 1: 2	{96,192,336,720}	Weather
ECL	321	Hourly	26 304	7: 1: 2	{96,192,336,720}	Electricity
Exchange	8	Daily	7588	7: 1: 2	{96,192,336,720}	Economy
Solar-Energy	137	10 min	52 560	7: 1: 2	{96,192,336,720}	Energy

Table 2

Statistics of the PEMS traffic flow benchmarks, including dimensionality, 5-minute sampling frequency, sequence length, data split, and prediction horizons.

Dataset	Dim	Frequency	Total len.	Split	Prediction len.	Information
PEMS03	358	5 min	26 209	6: 2: 2	{12,24,48,96}	Transportation
PEMS04	307	5 min	16 992	6: 2: 2	{12,24,48,96}	Transportation
PEMS07	883	5 min	28 224	6: 2: 2	{12,24,48,96}	Transportation
PEMS08	170	5 min	17 856	6: 2: 2	{12,24,48,96}	Transportation

and DLinear [21]. This selection strategy ensures fair and meaningful comparisons across different types of multivariate time series.

Setup and Evaluation. All methods adopt the same prediction horizons $H \in \{96, 192, 336, 720\}$ with a fixed look-back [35] window length of $L = 96$. All implementations are based on PyTorch and are executed on a single NVIDIA GeForce RTX 4090 GPU with 24 GB memory [48]. Models are trained using the Adam optimizer for 30 epochs [49], with early stopping applied on the validation set using a patience of 5. The initial learning rate is set to 2×10^{-3} for most datasets and reduced to 1×10^{-3} for smaller datasets (e.g., the ETT series). The batch size is fixed at 32. Model performance is evaluated using Mean Squared Error (MSE) and Mean Absolute Error (MAE).

Table 3

Full results on eight datasets. The best results are shown in **bold**, while the second-best results are underlined. The look-back window size is $L = 96$, and the prediction horizons are $H \in \{96, 192, 336, 720\}$. ‘Avg’ denotes the average results across all prediction horizons.

Models	PARCNet (Ours)		TQNet (2025)		TimePro (2025)		Amplifier (2025)		iTransformer (2024)		SOFTS (2024)		Fredformer (2024)		TiDE (2024)		TimeNet (2024)		PatchTST (2023)		
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
ETTm1	96	0.303	0.332	<u>0.311</u>	<u>0.353</u>	0.326	0.364	0.316	0.355	0.334	0.368	0.325	0.361	0.326	0.361	0.364	0.387	0.338	0.375	0.329	0.365
	192	<u>0.357</u>	0.360	0.356	<u>0.378</u>	0.367	0.383	0.361	0.381	0.377	0.391	0.375	0.389	0.363	0.380	0.398	0.404	0.374	0.387	0.380	0.394
	336	0.388	0.383	<u>0.390</u>	<u>0.401</u>	0.402	0.409	0.393	0.402	0.426	0.420	0.405	0.412	0.395	0.403	0.428	0.425	0.410	0.411	0.400	0.410
	720	0.457	0.422	0.452	<u>0.440</u>	0.469	0.446	0.455	0.440	0.491	0.459	0.466	0.447	<u>0.453</u>	<u>0.438</u>	0.487	0.461	0.478	0.450	0.475	0.453
	avg	0.376	0.374	<u>0.377</u>	<u>0.393</u>	0.391	0.400	0.381	0.394	0.407	0.410	0.393	0.403	0.384	0.395	0.419	0.419	0.400	0.406	0.396	0.406
ETTm2	96	0.167	0.244	<u>0.173</u>	<u>0.256</u>	0.178	0.260	0.176	0.258	0.180	0.264	0.180	0.261	0.177	0.259	0.207	0.305	0.187	0.267	0.184	0.264
	192	0.231	0.287	<u>0.238</u>	<u>0.298</u>	0.242	0.303	0.239	0.300	0.250	0.309	0.246	0.306	0.241	0.300	0.290	0.364	0.249	0.309	0.246	0.306
	336	0.291	0.326	0.301	0.340	0.303	0.342	<u>0.297</u>	<u>0.338</u>	0.311	0.348	0.319	0.352	0.302	0.340	0.377	0.422	0.321	0.351	0.308	0.346
	720	<u>0.394</u>	0.387	0.397	<u>0.396</u>	0.400	0.399	0.393	0.396	0.412	0.407	0.405	0.401	0.397	0.396	0.558	0.524	0.408	0.403	0.409	0.402
	avg	0.271	0.311	0.277	<u>0.323</u>	0.281	0.326	<u>0.276</u>	0.323	0.288	0.332	0.287	0.330	0.279	0.324	0.358	0.404	0.291	0.333	0.287	0.330
ETTh1	96	0.368	0.382	<u>0.371</u>	0.393	0.375	0.398	0.371	<u>0.392</u>	0.386	0.405	0.381	0.399	0.373	0.392	0.479	0.464	0.384	0.402	0.394	0.406
	192	<u>0.427</u>	0.413	0.428	0.426	0.427	0.429	0.426	0.422	0.441	0.436	0.435	0.431	0.433	<u>0.420</u>	0.525	0.492	0.436	0.429	0.440	0.435
	336	<u>0.467</u>	0.434	0.476	0.446	0.472	0.450	0.448	0.434	0.487	0.458	0.480	0.452	0.470	<u>0.437</u>	0.565	0.515	0.491	0.469	0.491	0.462
	720	<u>0.472</u>	0.454	0.487	0.470	0.476	0.474	0.476	0.464	0.503	0.491	0.499	0.488	0.467	<u>0.456</u>	0.594	0.558	0.521	0.500	0.487	0.479
	avg	<u>0.434</u>	0.421	0.441	0.434	0.438	0.438	0.430	0.428	0.454	0.447	0.449	0.442	0.435	<u>0.426</u>	0.541	0.507	0.458	0.450	0.453	0.446
ETTh2	96	<u>0.283</u>	0.328	0.295	0.343	0.293	0.345	0.279	<u>0.337</u>	0.297	0.349	0.297	0.347	0.293	0.342	0.400	0.440	0.340	0.374	0.288	0.340
	192	<u>0.364</u>	0.379	0.367	0.393	0.367	0.394	0.359	<u>0.389</u>	0.380	0.400	0.373	0.394	0.371	0.389	0.528	0.509	0.402	0.414	0.376	0.395
	336	0.406	0.413	0.417	0.427	0.419	0.431	0.377	0.406	0.428	0.432	0.410	0.426	<u>0.382</u>	<u>0.409</u>	0.643	0.571	0.452	0.452	0.440	0.451
	720	0.414	0.430	0.433	0.446	0.427	0.445	0.420	<u>0.432</u>	0.427	0.445	0.411	0.433	0.415	0.434	0.874	0.679	0.462	0.468	0.436	0.453
	avg	0.367	0.388	0.378	0.402	0.377	0.403	0.359	<u>0.391</u>	0.383	0.407	0.373	0.400	<u>0.365</u>	0.393	0.611	0.550	0.414	0.427	0.385	0.410
ECL	96	0.134	0.225	0.134	<u>0.229</u>	<u>0.139</u>	0.234	0.147	0.243	0.148	0.240	0.143	0.233	0.147	0.241	0.237	0.329	0.168	0.272	0.164	0.251
	192	0.152	0.242	<u>0.154</u>	<u>0.247</u>	<u>0.156</u>	0.249	0.157	0.251	0.162	0.253	0.158	0.248	0.165	0.258	0.236	0.330	0.184	0.289	0.173	0.262
	336	0.168	0.261	<u>0.169</u>	<u>0.264</u>	0.172	0.267	0.174	0.269	0.178	0.269	0.178	0.269	0.177	0.273	0.249	0.344	0.198	0.300	0.190	0.279
	720	0.201	0.289	0.201	<u>0.294</u>	0.209	0.299	<u>0.206</u>	0.296	0.225	0.317	0.218	0.305	0.213	0.304	0.284	0.373	0.220	0.320	0.230	0.313
	avg	0.164	0.254	0.164	<u>0.259</u>	<u>0.169</u>	0.262	0.171	0.265	0.178	0.270	0.174	0.264	0.175	0.269	0.251	0.344	0.192	0.295	0.189	0.276
Exchange	96	0.081	0.197	0.084	0.203	0.085	0.204	<u>0.083</u>	<u>0.202</u>	0.086	0.206	0.091	0.209	0.084	0.202	0.094	0.218	0.107	0.234	0.088	0.205
	192	0.171	0.293	0.177	<u>0.297</u>	0.178	0.299	<u>0.175</u>	0.297	0.177	0.299	0.176	0.303	0.183	0.302	0.184	0.307	0.226	0.344	0.176	0.299
	336	<u>0.326</u>	0.410	0.329	0.418	0.328	0.414	0.328	0.414	0.331	0.417	0.329	0.416	0.335	0.418	0.349	0.431	0.367	0.448	0.301	0.397
	720	<u>0.826</u>	0.690	0.831	0.698	0.817	0.679	0.858	0.696	0.847	0.691	0.848	<u>0.680</u>	0.893	0.711	0.852	0.698	0.964	0.746	0.901	0.714
	avg	0.351	0.398	0.355	0.404	<u>0.352</u>	<u>0.399</u>	0.361	0.402	0.360	0.403	0.361	0.402	0.374	0.408	0.370	0.413	0.416	0.443	0.367	0.404
Weather	96	0.148	0.183	0.157	<u>0.200</u>	0.166	0.207	<u>0.156</u>	0.204	0.174	0.214	0.166	0.208	0.163	0.207	0.202	0.261	0.172	0.220	0.176	0.217
	192	0.194	0.230	<u>0.206</u>	<u>0.245</u>	0.216	0.254	0.209	0.249	0.221	0.254	0.217	0.253	0.211	0.251	0.242	0.298	0.219	0.261	0.221	0.256
	336	0.252	0.274	<u>0.262</u>	<u>0.287</u>	0.273	0.296	0.264	0.290	0.278	0.296	0.282	0.300	0.267	0.292	0.287	0.335	0.280	0.306	0.275	0.296
	720	0.333	0.328	0.344	0.342	0.351	0.346	<u>0.343</u>	0.342	0.358	0.347	0.356	0.351	0.343	<u>0.341</u>	0.351	0.386	0.365	0.359	0.352	0.346
	avg	0.232	0.254	<u>0.242</u>	<u>0.269</u>	0.251	0.276	0.243	0.271	0.258	0.278	0.255	0.278	0.246	<u>0.272</u>	0.271	0.320	0.259	0.287	0.256	0.279
Solar	96	0.173	0.189	0.173	0.233	0.196	0.237	0.193	0.232	0.203	0.237	0.200	<u>0.230</u>	<u>0.185</u>	0.233	0.312	0.399	0.373	0.358	0.265	0.323
	192	<u>0.217</u>	0.217	0.199	<u>0.257</u>	0.231	0.263	0.228	0.259	0.233	0.261	0.229	<u>0.253</u>	<u>0.227</u>	0.253	0.339	0.416	0.397	0.376	0.288	0.332
	336	<u>0.232</u>	0.231	0.211	<u>0.263</u>	0.250	0.281	0.259	0.281	0.248	0.273	0.243	0.269	0.246	0.284	0.368	0.430	0.420	0.380	0.301	0.339
	720	<u>0.242</u>	0.236	0.209	<u>0.270</u>	0.253	0.285	0.249	0.278	0.249	0.275	0.245	0.272	0.247	0.276	0.370	0.425	0.420	0.381	0.295	0.336
	avg	<u>0.216</u>	0.218	0.198	<u>0.256</u>	0.232	0.266	0.233	0.264	0.233	0.262	0.229	0.256	0.226	0.262	0.347	0.417	0.403	0.374	0.287	0.333
1st Count	22	37	6	0	1	1	8	1	0	0	1	0	1	0	0	0	0	0	0	1	1

5.2. Main result

In this section, we compare the proposed PARCNet with several state-of-the-art baseline methods on eight standard time series forecasting datasets. The experimental results are reported in Table 3. Overall, PARCNet exhibits competitive and consistently strong performance across most datasets and prediction horizons, achieving 22 best MSE and 37 best MAE results. Under the average prediction horizon setting, PARCNet attains five best MSE results and all best MAE results across the eight datasets, indicating its favorable overall performance and stability.

In particular, PARCNet shows notable improvements on the Weather and ETTh2 datasets. On the Weather dataset, compared with existing state-of-the-art models, PARCNet achieves average reductions of 4.64% in MSE and 5.81% in MAE. On the ETTh2 dataset, the corresponding reductions are 2.98% in MSE and 3.70% in MAE. These results suggest that PARCNet maintains robust performance across different forecasting horizons and demonstrates good generalization capability in long-horizon time series forecasting tasks.

Error Distribution Analysis. Although MSE and MAE are widely used in time series forecasting, they emphasize different aspects of prediction errors. MSE is more sensitive to large errors due to the squared residuals, whereas MAE reflects the average absolute deviation more uniformly. Therefore, a model may achieve the best MSE but

Table 4

Quantile statistics of absolute prediction errors on representative settings. Q50, Q75, Q90, and Q95 denote the 50th, 75th, 90th, and 95th percentiles of absolute errors, respectively. Lower values indicate better performance. The best results are shown in bold, and the second-best results are underlined.

Dataset	Horizon	Model	Q50	Q75	Q90	Q95
ETTh1	96	PatchTST	<u>0.251</u>	<u>0.472</u>	0.954	1.342
		iTransformer	0.254	0.486	<u>0.886</u>	<u>1.333</u>
		PARCNet	0.247	0.469	0.835	1.291
Weather	96	PatchTST	<u>0.098</u>	0.275	0.555	<u>0.779</u>
		iTransformer	0.099	<u>0.272</u>	<u>0.545</u>	0.785
		PARCNet	0.077	0.235	0.471	0.683
Solar	96	PatchTST	<u>0.066</u>	<u>0.364</u>	<u>0.775</u>	<u>1.094</u>
		iTransformer	0.069	0.396	0.922	1.322
		PARCNet	0.014	0.241	0.702	1.008

not necessarily the best MAE in some cases, or vice versa. To further examine the stability of PARCNet from the perspective of error distribution, we compute quantile statistics of absolute prediction errors, including Q50, Q75, Q90, and Q95, on representative datasets under the prediction horizon of 96.

As shown in Table 4, PARCNet achieves the best quantile errors across all evaluated settings. In particular, the lower Q90 and Q95

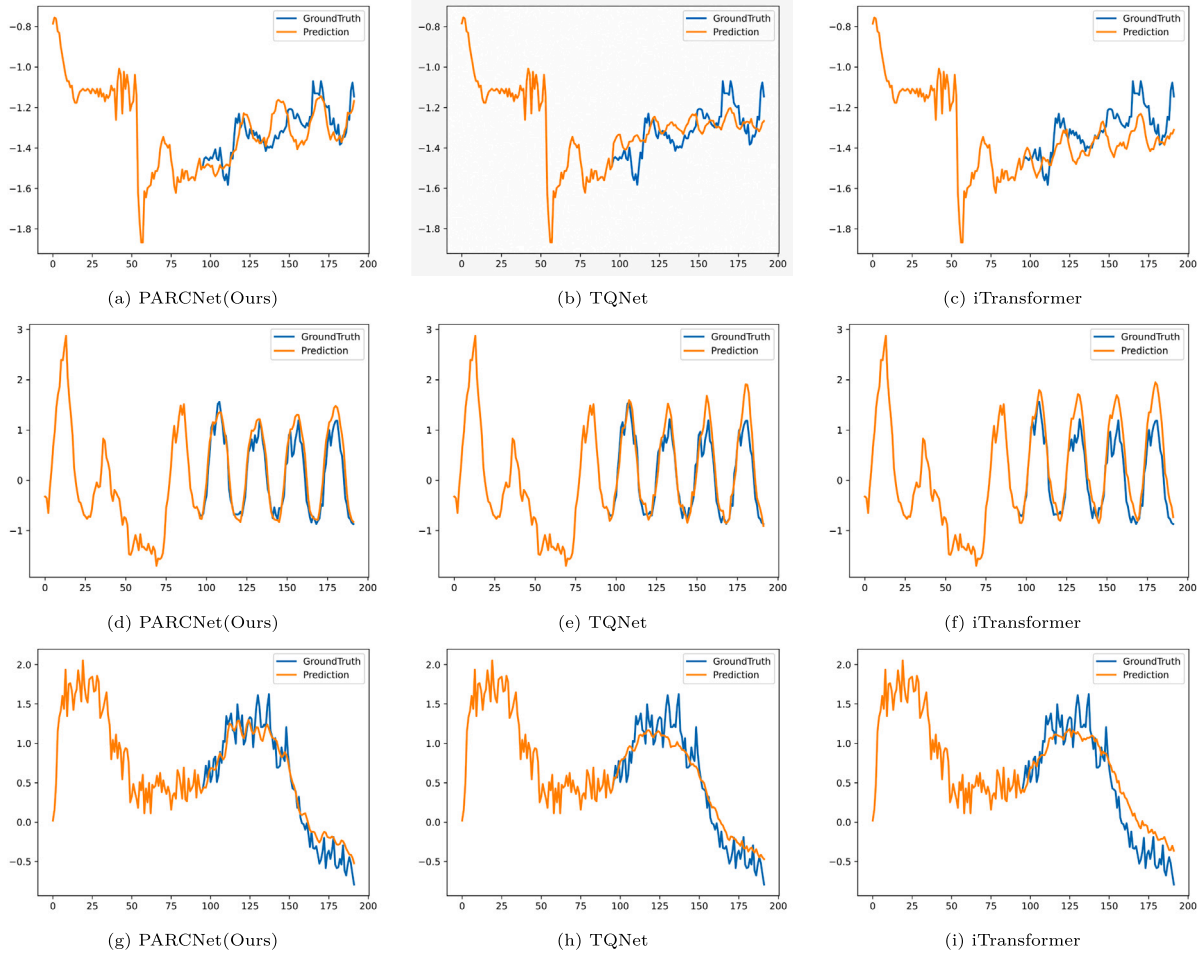


Fig. 4. Forecasting visualizations on three datasets. Rows correspond to datasets: ETTh1 (top), ECL (middle), and PEMS03 (bottom). Columns correspond to models: PARCNet (ours) (left), TQNet (middle), and iTransformer (right). In each subplot, the ground truth is compared with the predicted sequence.

values indicate that PARCNet can effectively suppress large local deviations and improve tail stability. These results provide a more comprehensive explanation for the occasional mismatch between MSE and MAE rankings in Table 3: while MSE and MAE emphasize different regions of the error distribution, PARCNet maintains stable performance not only in average errors but also in the upper-tail error region.

Furthermore, on the more challenging traffic flow datasets, including PEMS03, PEMS04, PEMS07, and PEMS08, PARCNet also achieves consistently favorable results. As shown in Table 5, the proposed model obtains 20 best MSE and 19 best MAE results across these four datasets, outperforming the baseline methods in most settings. Specifically, on the PEMS03 dataset, PARCNet reduces MSE and MAE by 35.4% and 20.7%, respectively; on PEMS04, by 24.6% and 17.6%; on PEMS07, by 35.8% and 27.0%; and on PEMS08, by 32.5% and 21.6%.

As shown in Fig. 4, we provide a visual comparison of the forecasting results across the different datasets, offering an intuitive view of the model's predictive performance. These results further indicate the effectiveness of PARCNet in modeling complex spatio-temporal dynamics and its applicability across different traffic forecasting scenarios.

5.3. Model analysis

Model efficiency. In Fig. 5, we compare the efficiency of different models on two representative datasets: Weather (21 variables) and Solar-Energy (137 variables), both with a lookback length of 96. Overall, PARCNet achieves a favorable trade-off between accuracy and efficiency on both datasets, maintaining competitive forecasting accuracy while achieving excellent training speed and memory efficiency.

Specifically, on datasets with relatively fewer variables (Weather), PARCNet exhibits higher efficiency than other Transformer-based models. On datasets with a larger number of variables (Solar-Energy), PARCNet continues to be competitive with Transformer-based counterparts in both speed and stability, with only slightly higher memory usage (in GB) compared to the linear model DLinear. These results indicate that PARCNet achieves a superior balance between computational efficiency and predictive performance across multivariate scenarios of different scales.

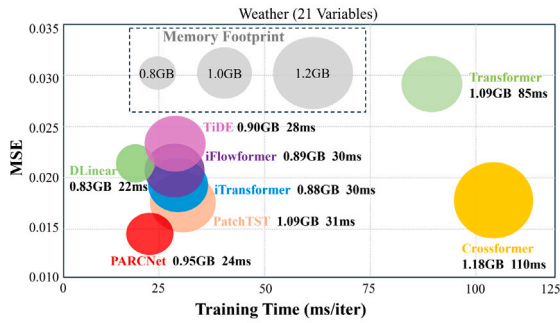
Attention Structure Visualization. Based on the visualization of attention weights on the Weather dataset in Fig. 6, we analyze the impact of SFM on attention modeling from two complementary perspectives, namely two-dimensional heatmaps and three-dimensional surface plots. Since the query, key, and value representations in subsequent attention mechanisms are all derived from the feature representation produced by SFM, the structural adjustment introduced by this module is reflected in the distribution patterns of the attention matrices.

From the two-dimensional attention heatmaps, it can be observed that incorporating SFM leads to a smoother and more continuous distribution of attention weights across variables. While key inter-variable interactions remain clearly emphasized, the remaining regions exhibit more uniform and fine-grained transitions, without obvious fragmented low-response areas. This indicates that the model is able to strengthen critical feature interactions while maintaining stable awareness of auxiliary variable relationships. In contrast, the attention matrices obtained without SFM display more dispersed high- and low-response regions, suggesting a less balanced modeling of inter-variable dependencies.

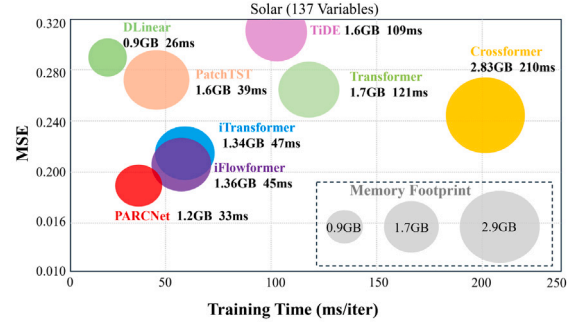
Table 5

Full results of the PEMS forecasting task. We compare extensive competitive models under different prediction lengths following the setting of OLinear. The input length is set to 96 for all baselines. Avg means the average results from all four prediction lengths.

Models	PARCNet (Ours)		TimeMixer++ (2025)		TimeMixer (2025)		FilterNet (2024)		FITS (2024)		CARD (2024)		Fredformer (2024)		Crossformer (2023)		DLinear (2023)		
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
PEMS03	12	0.059	0.159	0.097	0.208	0.076	0.188	0.071	0.177	0.117	0.226	0.072	0.177	<u>0.068</u>	<u>0.174</u>	0.090	0.203	0.122	0.243
	24	0.078	0.181	0.120	0.230	0.113	0.226	0.102	0.213	0.235	0.324	0.107	0.217	<u>0.093</u>	<u>0.202</u>	0.121	0.240	0.201	0.317
	48	0.114	0.220	0.170	0.272	0.191	0.292	0.162	0.272	0.541	0.521	0.194	0.302	<u>0.146</u>	<u>0.258</u>	0.202	0.319	0.333	0.425
	96	0.179	0.278	0.274	0.342	0.288	0.363	0.244	0.340	1.062	0.790	0.323	0.402	<u>0.228</u>	<u>0.330</u>	0.262	0.367	0.457	0.515
	avg	0.108	0.210	0.165	0.263	0.167	0.267	0.145	0.251	0.489	0.465	0.174	0.275	<u>0.135</u>	<u>0.243</u>	0.169	0.282	0.278	0.375
PEMS04	12	0.069	0.168	0.099	0.214	0.092	0.204	<u>0.082</u>	0.190	0.129	0.239	0.089	0.194	0.085	<u>0.189</u>	0.095	0.218	0.148	0.272
	24	0.083	0.186	0.115	0.231	0.128	0.243	<u>0.110</u>	<u>0.224</u>	0.246	0.337	0.128	0.234	0.117	0.224	0.131	0.256	0.224	0.340
	48	0.108	0.214	<u>0.144</u>	<u>0.261</u>	0.213	0.315	0.160	0.276	0.458	0.539	0.224	0.321	0.174	0.276	0.205	0.326	0.355	0.437
	96	0.157	0.263	<u>0.185</u>	<u>0.297</u>	0.307	0.384	0.234	0.343	1.181	0.843	0.382	0.445	0.273	0.354	0.402	0.457	0.452	0.504
	avg	0.104	0.208	<u>0.136</u>	<u>0.251</u>	0.185	0.287	0.146	0.258	0.531	0.489	0.206	0.299	0.162	0.261	0.209	0.314	0.295	0.388
PEMS07	12	0.054	0.144	0.090	0.197	0.073	0.184	0.064	0.163	0.109	0.222	0.068	0.166	<u>0.063</u>	<u>0.158</u>	0.094	0.200	0.115	0.242
	24	0.072	0.164	0.110	0.219	0.111	0.219	0.093	0.200	0.230	0.327	0.103	0.206	<u>0.089</u>	<u>0.192</u>	0.139	0.247	0.210	0.329
	48	0.107	0.198	0.149	0.256	0.237	0.328	0.137	0.248	0.551	0.531	0.165	0.268	<u>0.136</u>	<u>0.241</u>	0.311	0.369	0.398	0.458
	96	0.154	0.240	0.258	0.359	0.303	0.354	0.198	0.306	1.112	0.809	0.258	0.346	<u>0.197</u>	<u>0.298</u>	0.396	0.442	0.594	0.553
	avg	0.097	0.187	0.152	0.258	0.181	0.271	0.123	0.229	0.500	0.472	0.149	0.247	<u>0.121</u>	<u>0.222</u>	0.235	0.315	0.329	0.395
PEMS08	12	0.070	0.167	0.119	0.222	0.091	0.201	<u>0.080</u>	0.182	0.122	0.233	0.080	<u>0.181</u>	0.081	0.185	0.165	0.214	0.154	0.276
	24	0.092	0.189	0.149	0.249	0.137	0.246	<u>0.114</u>	0.219	0.236	0.330	0.118	0.220	<u>0.112</u>	<u>0.214</u>	0.215	0.260	0.248	0.353
	48	0.144	0.233	0.206	0.292	0.265	0.343	0.184	0.284	0.562	0.540	0.199	0.289	<u>0.174</u>	<u>0.267</u>	0.315	0.355	0.440	0.470
	96	0.261	0.294	0.329	0.355	0.410	0.407	0.309	0.256	1.216	0.846	0.405	0.431	<u>0.277</u>	<u>0.335</u>	0.377	0.397	0.674	0.565
	avg	0.142	0.221	0.200	0.279	0.226	0.299	0.172	0.260	0.534	0.487	0.201	0.280	<u>0.161</u>	<u>0.250</u>	0.268	0.307	0.379	0.416
1st Count	20	19	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0

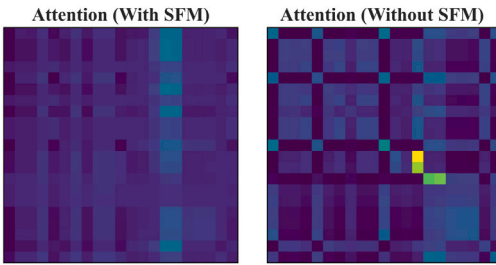


(a) Weather

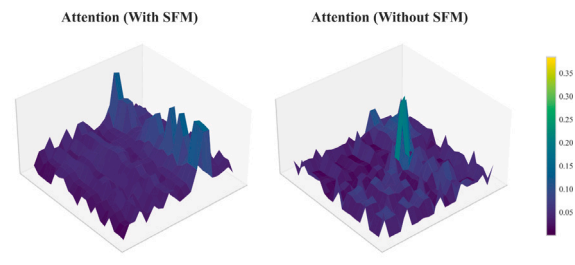


(b) Solar-Energy

Fig. 5. Model efficiency comparison on Weather (21 variables) and Solar-Energy (137 variables) under the input-96/predict-96 setting. The x-axis denotes training time (ms/iter) and the y-axis denotes MSE; bubble size indicates memory footprint (GB).



(a) 2D attention



(b) 3D attention

Fig. 6. Attention structure visualization on the Weather dataset. (a) 2D attention heatmap; (b) 3D attention surface. Compared with the model without SFM, incorporating SFM yields a more continuous and smoother distribution of attention weights, with fewer spike-like responses and a stronger focus on key correlated regions, suggesting that SFM helps build more stable cross-variable dependency modeling.

The three-dimensional attention surface plots serve as a spatial complement to the two-dimensional observations and further support these findings. With SFM, the attention matrices exhibit a more continuous and smooth spatial structure, where high-response regions appear as locally aggregated areas rather than isolated sharp peaks, and the remaining regions show stable variations. Without SFM, the attention distributions tend to present coexisting local protrusions and relatively flat regions. These observations suggest that SFM facilitates

the formation of more balanced and structured interaction patterns within the attention mechanism.

Mechanism Analysis of the SFM Module. In multivariate time series forecasting, standard attention-based [29] modeling often exhibits an implicit amplitude or energy bias [50]. Variables with larger magnitudes or stronger short-term fluctuations are more likely to dominate the attention allocation. While this mechanism helps capture strong signals, it may also cause attention to concentrate excessively on a few

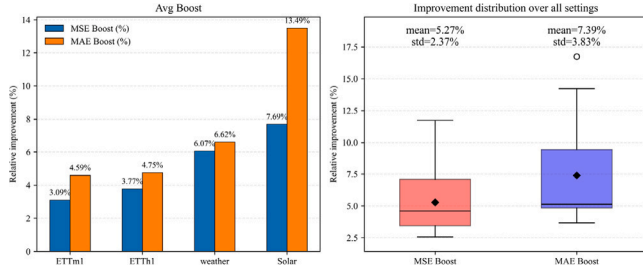


Fig. 7. Summary of SFM gains across datasets and settings. The bar chart reports the average relative error reduction (%) in MSE and MAE (i.e., *Boost* in Table 6) after incorporating SFM, while the boxplots show the distribution of improvements across all prediction horizons and datasets, indicating consistent benefits.

channels in non-stationary environments [51]. As a result, variables that are structurally correlated but have relatively smaller amplitudes can be overlooked, which may lead to cross-variable misfusion and unstable long-horizon predictions.

The SFM addresses this issue by explicitly introducing amplitude and phase information to provide structural guidance for feature modulation [52]. Unlike amplitude, which mainly reflects signal intensity, phase information describes the relative evolution and synchronization relationships among variables [53]. In many oscillatory systems, phase consistency across variables is often more stable than amplitude correlation for revealing underlying coupling structures [54]. By extracting amplitude and phase features from the analytic signal and using them to generate a modulation signal, SFM performs structure-aware correction on the original representations, enabling the model to highlight dominant variables while still preserving the contributions of phase-related variables [55,56].

Consequently, the attention distribution becomes less dominated by local amplitude variations and more guided by the oscillatory structural consistency among variables [57–59]. Empirically, the attention maps after introducing SFM exhibit a more balanced yet still discriminative pattern: key variables remain emphasized, while phase-related variables receive more stable attention. This structure-aware modulation helps mitigate attention collapse and improves the robustness of multivariate modeling, particularly for long-horizon forecasting tasks [17, 60,61].

5.4. Ablation study

Ablation Study of Core Modules. This section evaluates the effectiveness of the proposed Structure-aware Feature Module (SFM). Ablation studies are conducted on four representative benchmark datasets, including ETTm1, ETTh1, Weather, and Solar, by comparing the forecasting performance of the model with and without the SFM to quantify its contribution.

As shown in Table 6, incorporating SFM consistently leads to stable and consistent performance improvements across all datasets, with particularly notable gains on the Solar dataset, where MSE and MAE are reduced by approximately 7.92% and 13.4%, respectively. Fig. 7 provides a visualization of the average performance gains (Boost), offering an intuitive illustration of the stable improvements achieved by SFM across different datasets and prediction horizons, and serving as a complementary verification of the quantitative results reported in Table 6. These results indicate that SFM enhances the quality of input representations through gated modulation and residual correction, effectively suppressing redundant and unstable components. As a result, the features fed into subsequent attention modules become more stable and discriminative, thereby improving the overall forecasting performance.

Table 6

Ablation results of SFM on four benchmark datasets. MSE/MAE are reported for horizons $H \in \{96, 192, 336, 720\}$ and their average. *Boost* denotes the relative error reduction (%) brought by incorporating SFM compared with w/o SFM.

Dataset		MSE			MAE		
		w/o SFM	SFM	Boost	w/o SFM	SFM	Boost
ETTh1	96	0.313	0.303	3.195%	0.349	0.332	4.871%
	192	0.371	0.357	3.744%	0.379	0.360	5.013%
	336	0.401	0.388	3.242%	0.402	0.383	4.726%
	720	0.469	0.457	2.559%	0.438	0.422	3.653%
	avg	0.388	0.376	3.093%	0.392	0.374	4.592%
ETTh1	96	0.387	0.368	4.910%	0.402	0.382	4.975%
	192	0.447	0.427	4.474%	0.435	0.413	5.057%
	336	0.484	0.467	3.512%	0.455	0.434	4.615%
	720	0.486	0.472	2.881%	0.474	0.454	4.219%
	avg	0.451	0.434	3.769%	0.442	0.421	4.751%
Weather	96	0.159	0.148	6.918%	0.201	0.183	8.955%
	192	0.203	0.194	4.433%	0.245	0.230	6.122%
	336	0.273	0.252	7.682%	0.296	0.274	7.432%
	720	0.351	0.333	5.128%	0.346	0.328	5.202%
	avg	0.247	0.232	6.073%	0.272	0.254	6.618%
Solar	96	0.196	0.173	11.735%	0.227	0.189	16.740%
	192	0.235	0.217	7.660%	0.253	0.217	14.229%
	336	0.251	0.232	7.570%	0.261	0.231	11.494%
	720	0.254	0.242	4.724%	0.265	0.236	10.943%
	avg	0.234	0.216	7.692%	0.252	0.218	13.492%

Table 7

Comparison of transformation choices for oscillatory feature extraction. MSE/MAE are reported for horizons $H \in \{96, 192, 336, 720\}$ and their average, by replacing the Hilbert-based analytic-signal extraction in SFM with Fourier- or wavelet-based alternatives under the same framework.

Dataset	Models	Ours		Fourier		Wavelet	
		MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.368	0.382	0.372	0.391	0.369	0.385
	192	0.427	0.413	0.435	0.421	0.430	0.417
	336	0.467	0.434	0.472	0.439	0.467	0.435
	720	0.472	0.454	0.481	0.459	0.471	0.453
	avg	0.434	0.421	0.440	0.427	0.434	0.422
Weather	96	0.148	0.183	0.155	0.191	0.150	0.186
	192	0.194	0.230	0.199	0.239	0.196	0.230
	336	0.252	0.274	0.261	0.281	0.253	0.273
	720	0.333	0.328	0.341	0.332	0.334	0.328
	avg	0.232	0.254	0.239	0.260	0.233	0.254
Solar	96	0.173	0.189	0.181	0.195	0.179	0.193
	192	0.217	0.217	0.220	0.221	0.219	0.220
	336	0.232	0.231	0.239	0.239	0.232	0.230
	720	0.242	0.236	0.250	0.241	0.244	0.237
	avg	0.216	0.218	0.222	0.224	0.218	0.220

Ablation on Frequency Transformation in SFM. We further investigate the impact of different frequency transformation methods used in the SFM module by replacing the Hilbert transform methods with Fourier and wavelet transforms. The results on ETTh1, Weather, and Solar datasets are summarized in Table 7. Overall, the proposed Hilbert-based design consistently achieves the lowest average MSE and MAE across all datasets and forecasting horizons. While Fourier and wavelet transforms yield competitive performance, they generally lead to higher errors, particularly under longer prediction horizons. These observations indicate that the Hilbert transform provides a more effective characterization of oscillatory structures in multivariate time series, contributing to more stable and accurate forecasting performance.

Ablation on Feature Combinations in SFGM. To separately examine the contributions of amplitude and phase information in the gating module, we conduct fine-grained ablation experiments on different amplitude-phase feature combinations. Specifically, we compare

Table 8

Sensitivity to feature composition in the gating module. Results under amplitude-only, phase-only, and mixed settings show that jointly using amplitude and phase features achieves the best performance, suggesting complementary contributions.

Dataset	Amp		Phase				Metric	
	logAx	logAt	$\cos \phi_x$	$\sin \phi_x$	$\cos \phi_t$	$\sin \phi_t$	MSE	MAE
ETTh1	✓	✓	×	×	×	×	0.375	0.387
	×	×	✓	✓	✓	✓	0.371	0.386
	✓	×	✓	✓	×	×	0.374	0.387
	✓	✓	✓	✓	✓	✓	0.368	0.382
ETTm1	✓	✓	×	×	×	×	0.307	0.335
	×	×	✓	✓	✓	✓	0.308	0.337
	✓	×	✓	✓	×	×	0.306	0.335
	✓	✓	✓	✓	✓	✓	0.303	0.332
Weather	✓	✓	×	×	×	×	0.153	0.191
	×	×	✓	✓	✓	✓	0.151	0.190
	✓	×	✓	✓	×	×	0.152	0.192
	✓	✓	✓	✓	✓	✓	0.148	0.183
Solar	✓	✓	×	×	×	×	0.181	0.199
	×	×	✓	✓	✓	✓	0.179	0.197
	✓	×	✓	✓	×	×	0.182	0.200
	✓	✓	✓	✓	✓	✓	0.173	0.189
ECL	✓	✓	×	×	×	×	0.141	0.232
	×	×	✓	✓	✓	✓	0.139	0.230
	✓	×	✓	✓	×	×	0.140	0.231
	✓	✓	✓	✓	✓	✓	0.134	0.225

Table 9

Ablation of structure-aware residual correction (SRC) formulations. Different fusion strategies are evaluated, and the SRC-style update $f = X + g \odot (T - X)$ achieves the best overall accuracy across datasets in terms of MSE/MAE.

Dataset	ETTh1		Weather		Solar	
	MSE	MAE	MSE	MAE	MSE	MAE
$f = X$	0.374	0.391	0.155	0.194	0.190	0.201
$f = X + T$	0.370	0.387	0.151	0.188	0.188	0.196
$f = X + g \odot T$	0.371	0.385	0.148	0.185	0.176	0.192
$f = g \odot X + g \odot T$	0.369	0.384	0.149	0.184	0.174	0.190
$f = X + g \odot (T - X)$	0.368	0.382	0.148	0.183	0.173	0.189

amplitude-only features, phase-only features, partial amplitude-phase combinations, and the full amplitude-phase setting. As shown in Table 8, both amplitude-only and phase-only configurations provide useful structural information, while the complete combination consistently achieves the best performance across all evaluated datasets. These results indicate that amplitude and phase provide complementary structural information: amplitude describes local oscillation intensity, whereas phase captures relative evolution and synchronization patterns.

Ablation on Structure-aware Residual Correction (SRC). To empirically justify the contribution of the learnable structural prior and the design of the proposed SRC formulation, we conduct ablation experiments on the ETTh1, Weather, and Solar datasets by replacing Eq. (15) with several alternative combination strategies.

As shown in Table 9, the proposed SRC formulation achieves the best overall MSE and MAE across the evaluated datasets. These results indicate that both the discrepancy direction $(T - X)$ and the adaptive gate g are important for stable feature correction. Direct addition or ungated fusion cannot provide the same balance between preserving original features and introducing structural guidance, which empirically supports the design choice of Eq. (15).

Ablation on SFM as a Plug-and-Play Module. To evaluate the generalizability of the SFM module, we integrate it as a plug-and-play component into three base models, iTransformer (2024) [32],

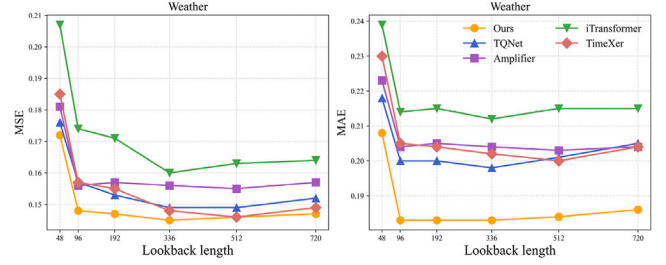


Fig. 8. Lookback-length analysis on Weather. Performance curves (MSE/MAE) versus lookback length highlight the robustness of PARCNet across different input window sizes.

PatchTST (2023) [3], and DLinear (2023) [21], and conduct experiments on the ETTh1, ETTh2, Weather, and Solar datasets. As shown in Table 10, incorporating SFM generally improves the forecasting performance of different backbones in most settings, especially in terms of average results across prediction horizons. Although a few individual horizons show comparable or slightly fluctuating results, the overall trend demonstrates that SFM can effectively enhance the representation capability of diverse forecasting models.

Efficiency Overhead of Plug-and-play SFM. Although Table 10 demonstrates that SFM generally improves different forecasting backbones in most settings, its plug-and-play usage also requires evaluating the additional computational cost. Therefore, in addition to the memory analysis in Fig. 5, we further report the parameter count and training time introduced by inserting SFM into representative forecasting backbones.

As shown in Table 11, SFM introduces only a very small number of additional trainable parameters. For example, the parameter count increases from 1.6779M to 1.6785M for iTransformer, from 4.2871M to 4.2876M for PatchTST, and from 0.0186M to 0.0189M for DLinear. This indicates that the additional parameter overhead of SFM is negligible. In terms of training time, SFM increases the per-epoch cost because it introduces amplitude-phase representation and structure-aware gating operations. Nevertheless, the absolute training time remains within a manageable range on all evaluated backbones. Considering the consistent accuracy improvements reported in Table 10, these results show that SFM can serve as an effective plug-and-play module with limited parameter overhead and acceptable training cost.

5.5. Hyperparameter sensitivity

Lookback Length Analysis. Fig. 8 illustrates the performance of the model on the Weather dataset in terms of MSE and MAE under different lookback window lengths. It can be observed that as the window length increases from 48 to 336, the errors of the evaluated methods generally decrease, indicating that appropriately increasing the historical information helps capture long-horizon dependencies in the time series. When the window is further extended to 512 and 720, the errors tend to stabilize or slightly increase, suggesting that excessively long input windows may introduce additional noise, thereby increasing the risk of overfitting during correlation modeling. Compared with other baseline models, the proposed method consistently achieves the best performance across all window lengths and is relatively insensitive to variations in the lookback length, demonstrating strong robustness.

Hyperparameters Analysis. This section presents a sensitivity analysis of the key hyperparameters in the 1D convolution layers of the SFGM module, including the number of hidden channels (hidden_channels) and the convolution kernel size (kernel_size). Experiments are conducted on the Weather dataset, and the predictive performance is evaluated in terms of MSE and MAE. As shown in Table 12, the results indicate that varying the number of hidden channels

Table 10

Plug-and-play evaluation of SFM. We insert SFM into three representative forecasters (iTransformer, PatchTST, and DLinear) and report MSE/MAE for the vanilla models (van.) and their SFM-augmented counterparts (+SFM) across datasets and prediction horizons.

Models		iTransformer (2024)				PatchTST (2023)				DLinear (2023)			
Metric		MSE		MAE		MSE		MAE		MSE		MAE	
Dataset	Horizon	van.	+SFM	van.	+SFM	van.	+SFM	van.	+SFM	van.	+SFM	van.	+SFM
ETTh1	96	0.386	0.386	0.405	0.400	0.394	0.384	0.406	0.395	0.386	0.385	0.400	0.398
	192	0.441	0.443	0.436	0.433	0.440	0.436	0.435	0.426	0.437	0.432	0.432	0.427
	336	0.487	0.482	0.458	0.453	0.491	0.476	0.462	0.446	0.481	0.473	0.459	0.454
	720	0.503	0.495	0.491	0.483	0.487	0.490	0.479	0.473	0.519	0.507	0.516	0.505
	avg	0.454	0.451	0.447	0.442	0.453	0.446	0.446	0.435	0.456	0.449	0.452	0.446
ETTh2	96	0.297	0.295	0.349	0.345	0.288	0.285	0.340	0.335	0.333	0.325	0.387	0.381
	192	0.380	0.372	0.400	0.395	0.376	0.365	0.395	0.387	0.477	0.456	0.476	0.464
	336	0.428	0.422	0.432	0.432	0.440	0.413	0.451	0.429	0.594	0.560	0.541	0.526
	720	0.427	0.429	0.445	0.447	0.436	0.428	0.453	0.450	0.831	0.828	0.657	0.649
	avg	0.383	0.379	0.407	0.404	0.385	0.372	0.410	0.400	0.559	0.542	0.515	0.505
Weather	96	0.174	0.168	0.214	0.207	0.176	0.164	0.217	0.209	0.196	0.191	0.255	0.242
	192	0.221	0.211	0.254	0.246	0.221	0.216	0.256	0.254	0.237	0.237	0.296	0.295
	336	0.278	0.269	0.296	0.281	0.275	0.269	0.296	0.292	0.283	0.273	0.335	0.314
	720	0.358	0.347	0.347	0.338	0.352	0.345	0.346	0.343	0.345	0.340	0.381	0.365
	avg	0.258	0.248	0.278	0.268	0.256	0.249	0.279	0.275	0.265	0.260	0.317	0.304
Solar	96	0.203	0.191	0.237	0.232	0.265	0.207	0.323	0.250	0.290	0.267	0.378	0.329
	192	0.233	0.228	0.261	0.259	0.288	0.244	0.332	0.276	0.320	0.296	0.398	0.346
	336	0.248	0.200	0.273	0.265	0.301	0.266	0.339	0.291	0.353	0.341	0.415	0.386
	720	0.249	0.231	0.275	0.269	0.295	0.256	0.336	0.281	0.356	0.344	0.413	0.394
	avg	0.233	0.222	0.262	0.256	0.287	0.243	0.333	0.275	0.330	0.312	0.401	0.363

Table 11

Parameter count and training time of representative forecasters with and without SFM on ETTh1 under the prediction horizon of 96. Params denotes the number of trainable parameters, and training time denotes the cost of one training epoch under the same experimental setting.

Backbone	Variant	Params (M)	Training time (s/epoch)
iTransformer	w/o SFM	1.6779	3.2932
	+SFM	1.6785	4.5437
PatchTST	w/o SFM	4.2871	3.0291
	+SFM	4.2876	5.1757
DLinear	w/o SFM	0.0186	2.3960
	+SFM	0.0189	3.9505

Table 12

Sensitivity analysis of SFGM convolution hyperparameters on the Weather dataset. MSE and MAE are reported under different numbers of hidden channels and convolution kernel sizes.

Channel	MSE	MAE	kernel_size	MSE	MAE
6	0.147	0.183	5	0.147	0.182
8	0.147	0.182	9	0.148	0.184
12	0.148	0.183	11	0.149	0.184
16	0.147	0.183	15	0.149	0.185
32	0.148	0.184	25	0.149	0.186

among 6, 8, 12, 16, and 32 yields generally stable performance, demonstrating good robustness. Notably, using 8 hidden channels achieves the best performance, with the lowest MSE and MAE, suggesting an optimal balance between feature representation capacity and computational efficiency.

To provide a more intuitive view, Fig. 9 presents boxplots for MSE and MAE under different learning rates, supplemented by the “win” and “rank” metrics. Although the performance differences among the learning rates are generally minor, the setting with a learning rate of 0.001 achieves the highest win rate and the lowest rank, suggesting

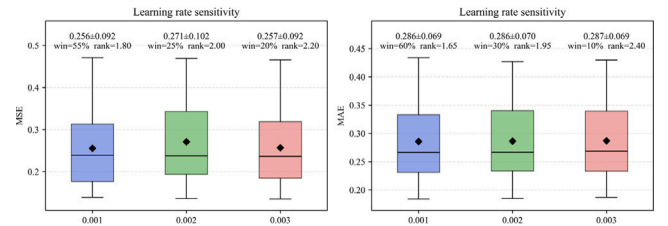


Fig. 9. Learning-rate sensitivity analysis. Boxplots of MSE (left) and MAE (right) under learning rates {0.001, 0.002, 0.003} across the evaluated forecasting settings. The annotated win and rank summarize the relative performance of each learning rate.

a slightly superior overall performance. Overall, this analysis demonstrates that the model is largely insensitive to the choice of learning rate within this range, while 0.001 can be considered a relatively optimal configuration.

Regarding the convolution kernel size, performance remains relatively stable as kernel_size increases from 5 to 25. Table 12 shows that a kernel size of 5 slightly outperforms other settings, achieving the minimum MSE and MAE. This indicates that a smaller kernel is more effective in capturing local patterns without excessive smoothing or introducing redundant information. Overall, the results suggest that the convolutional hyperparameters in SFGM have a limited impact on model performance within reasonable ranges, while the combination of 8 hidden channels and a kernel size of 5 provides the most favorable configuration for practical applications.

Random Seed Analysis. To evaluate the robustness of the proposed model with respect to initialization, experiments were conducted on five representative datasets, including ETTh1, ETTh2, Weather, Solar, and ECL, using different random seeds (2024, 2025, and 2026) [33]. For each seed, MSE and MAE were recorded across multiple prediction horizons, and the mean and standard deviation were computed to quantify performance variability.

Table 13

Random-seed sensitivity study across prediction horizons (96/192/336/720) on ETTm1, ETTm2, Weather, Solar, and ECL. Mean and Std quantify performance variability.

Dataset	Metric	2024		2025		2026		Mean		Std	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTM1	96	0.307	0.336	0.303	0.332	0.305	0.335	0.305	0.334	0.002	0.002
	192	0.361	0.366	0.357	0.360	0.359	0.364	0.359	0.363	0.002	0.003
	336	0.391	0.386	0.388	0.383	0.393	0.386	0.391	0.385	0.003	0.002
	720	0.464	0.428	0.457	0.422	0.352	0.426	0.424	0.425	0.002	0.003
ETTM2	96	0.166	0.244	0.167	0.244	0.170	0.248	0.168	0.245	0.002	0.002
	192	0.232	0.287	0.231	0.287	0.230	0.287	0.231	0.287	0.001	0.000
	336	0.292	0.326	0.291	0.326	0.300	0.335	0.294	0.329	0.005	0.005
	720	0.393	0.387	0.394	0.387	0.389	0.386	0.392	0.387	0.003	0.001
Weather	96	0.147	0.183	0.148	0.183	0.148	0.184	0.148	0.183	0.001	0.001
	192	0.196	0.230	0.194	0.230	0.195	0.230	0.195	0.230	0.001	0.000
	336	0.251	0.273	0.252	0.274	0.253	0.274	0.252	0.274	0.001	0.001
	720	0.331	0.327	0.333	0.328	0.331	0.326	0.332	0.327	0.001	0.001
Solar	96	0.175	0.189	0.173	0.189	0.181	0.200	0.176	0.193	0.004	0.006
	192	0.213	0.218	0.217	0.217	0.213	0.219	0.214	0.218	0.002	0.001
	336	0.234	0.231	0.232	0.231	0.235	0.232	0.234	0.231	0.002	0.001
	720	0.247	0.237	0.242	0.236	0.252	0.239	0.247	0.237	0.005	0.002
ECL	96	0.140	0.230	0.134	0.225	0.140	0.230	0.138	0.228	0.003	0.003
	192	0.157	0.245	0.152	0.242	0.157	0.245	0.155	0.244	0.003	0.002
	336	0.174	0.262	0.168	0.261	0.173	0.261	0.172	0.261	0.003	0.001
	720	0.203	0.286	0.201	0.289	0.205	0.287	0.203	0.287	0.002	0.002

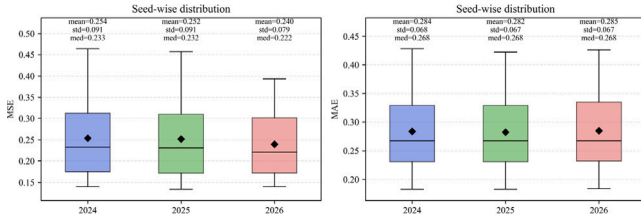


Fig. 10. Seed-wise robustness evaluation. Boxplots of MSE (left) and MAE (right) obtained under three random seeds (2024/2025/2026) across the evaluated forecasting settings. Mean, standard deviation, and median are reported above each box.

As summarized in Table 13, the model demonstrates consistently stable performance across all seeds and datasets. The standard deviations of both MSE and MAE remain low, indicating that the model is largely insensitive to random initialization and that the forecasting results are highly reproducible. While minor fluctuations are observed for certain longer prediction horizons, their magnitude is negligible relative to the overall error levels. Fig. 10 presents boxplots of MSE and MAE across different random seeds, providing an intuitive visualization that complements the table. In both metrics, the error distributions are tightly concentrated, further confirming the robustness and reliability of the proposed method under varying initializations.

Learning Rate Analysis. To investigate the sensitivity of the proposed model to the learning rate, we conducted hyperparameter experiments on five representative datasets, including ETTm1, ETTm2, Weather, Solar, and ECL. Three commonly used learning rates (0.001, 0.002, and 0.003) were compared across multiple prediction horizons, and MSE and MAE were recorded for each setting. The mean and standard deviation of the metrics were also computed to quantify performance variability. The results are summarized in Table 14. It can be observed that the model maintains stable forecasting performance across all learning rates, with minimal variations in both mean and standard deviation, indicating good robustness to this hyperparameter.

Additional Design Analysis. Beyond the sensitivity analysis of learning rate and convolution kernel size, we further clarify several design aspects of SFM. The dimension of the learnable structural prior T is not an independent hyperparameter; instead, T is defined to match the representation processed by SFM and is broadcast along the batch dimension during the forward pass. Moreover, PARCNet does not use a fixed feature retention ratio after the Hilbert transform. Different from frequency truncation or hard band-selection methods, the Hilbert-derived envelope and phase descriptors are retained at all time-channel positions and then adaptively weighted by the convolutional gate g . Therefore, feature usage is controlled in a data-driven soft-selection manner. For sampling-frequency adaptability, PARCNet does not rely on manually specified physical frequency bands or dataset-specific sampling intervals. The envelope and phase descriptors are computed on the discrete input sequence and further modulated by the learned time-channel adaptive gate, enabling the model to adapt to datasets with different sampling granularities. The consistent performance on ETT, Weather, ECL, Solar, and PEMS datasets also empirically supports this adaptability.

6. Conclusion

This paper proposes PARCNet (Phase-Aware Residual Correction Network) for multivariate time series forecasting. From a Hilbert analytic-signal perspective, PARCNet constructs aligned amplitude-phase descriptors and improves feature fusion via a structure-aware gating and residual correction mechanism, enabling more faithful modeling of local oscillatory states and enhanced cross-variable consistency. Extensive experiments across multiple real-world datasets and forecasting horizons show that PARCNet consistently outperforms state-of-the-art methods, delivering reliable gains in both accuracy and robustness, with particularly clear benefits in long-horizon scenarios. Future work will further characterize the operating conditions and theoretical properties of the proposed design (e.g., under multi-scale coupling or band overlap), and investigate multi-scale phase representations and lightweight implementations to improve generalization and deployment efficiency for high-dimensional and long-sequence settings.

Table 14

Effect of learning-rate choices on forecasting accuracy for ETTm1, ETTm2, Weather, Solar, and ECL under prediction horizons 96/192/336/720. Mean and Std summarize the stability across different learning-rate settings.

Dataset	Metric	0.001		0.002		0.003		Mean		Std	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTM1	96	0.306	0.339	0.314	0.344	0.314	0.344	0.311	0.342	0.005	0.003
	192	0.363	0.367	0.365	0.370	0.363	0.369	0.364	0.369	0.001	0.002
	336	0.401	0.392	0.400	0.393	0.397	0.393	0.399	0.393	0.002	0.001
	720	0.471	0.434	0.461	0.427	0.466	0.430	0.466	0.430	0.005	0.004
ETTM2	96	0.168	0.246	0.169	0.249	0.172	0.249	0.169	0.248	0.002	0.002
	192	0.244	0.299	0.234	0.291	0.240	0.295	0.239	0.295	0.005	0.004
	336	0.297	0.331	0.301	0.339	0.304	0.338	0.301	0.336	0.004	0.004
	720	0.386	0.384	0.403	0.401	0.394	0.394	0.394	0.393	0.009	0.009
Weather	96	0.148	0.184	0.148	0.185	0.151	0.187	0.149	0.185	0.002	0.002
	192	0.197	0.231	0.199	0.234	0.200	0.233	0.199	0.233	0.002	0.002
	336	0.250	0.272	0.253	0.273	0.255	0.276	0.253	0.274	0.003	0.002
	720	0.335	0.329	0.336	0.329	0.334	0.328	0.335	0.329	0.001	0.001
Solar	96	0.178	0.201	0.184	0.192	0.188	0.195	0.183	0.196	0.005	0.005
	192	0.214	0.216	0.220	0.220	0.218	0.222	0.217	0.219	0.003	0.003
	336	0.234	0.231	0.231	0.232	0.233	0.233	0.233	0.232	0.002	0.001
	720	0.245	0.238	0.242	0.237	0.246	0.240	0.244	0.238	0.002	0.002
ECL	96	0.139	0.230	0.136	0.227	0.135	0.225	0.137	0.227	0.002	0.003
	192	0.157	0.245	0.154	0.242	0.153	0.242	0.155	0.243	0.002	0.002
	336	0.173	0.261	0.173	0.261	0.174	0.261	0.173	0.261	0.001	0.000
	720	0.204	0.286	0.197	0.281	0.201	0.286	0.201	0.284	0.004	0.003

CRedit authorship contribution statement

Ziqiong Li: Methodology, Conceptualization. **Heyu Chai:** Writing – review & editing, Writing – original draft, Visualization. **Zhangyao Song:** Validation. **Shengjun Liu:** Validation, Supervision, Software, Funding acquisition, Data curation. **Xinru Liu:** Validation, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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